The Residual Core of the Collatz Problem

A Structural Decomposition

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Abstract

We identify the precise residual structure of the Collatz problem within the renewal framework of Kamanoi (2026a). The unsolved core is not probabilistic drift — the average Lyapunov exponent is negative, and this is rigorous. The residual is a single question: *can one rule out infinite arithmetic memory along a single orbit?*

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1. What Is Already Rigid (Level 0)

The companion paper (Kamanoi 2026a) establishes the following unconditionally:

The renewal system (X, σ, π) is exact.

Result	Content
Theorem EC	$\hat{\Phi}: \hat{\mathcal{A}} \cong X$ (2-adic homeomorphism)
Theorem 20-H	$\pi(k=j) = 1/2^j$ uniquely
Lemma S1	$P^N f = S^N f$ on $\ker E_\pi$
Theorem 25-I	$\mathrm{Cov}_\pi(k_0, k_N) = 2\cdot(1/2)^N$ exactly
Theorem IT2a	$\sum_{N=1}^\infty I_\pi(k_0;k_N) = 2\log 2$ exactly

The stationary renewal system carries a finite and explicitly computable total information memory, namely $2\log 2$ (the measure-theoretic entropy of one renewal step).

The $P^N f = S^N f$ identity is the structural fact: the renewal operator restricted to mean-

zero functions is *exactly* the shift-and-kill operator. No approximation is involved.

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2. What Divergence Must Violate (Level 1)

These results are rigorous.

Let $(n_i)_{i\geq0}$ be the odd Collatz iterates of n, with $v_i = v_2(3n_i+1)$.

Theorem D1-bwd. If the orbit diverges to $+\infty$: $$\liminf_{M\to\infty}\frac{1}{M}\sum_{i<M}v_i \leq \log_2 3.$$

Theorem NGM. If the orbit diverges: $\mu_n \neq \pi$.

Corollary. A divergent orbit cannot be π-typical. Under π: $E_\pi[v_i] = 2 > \log_2 3$. So a divergent orbit must systematically deviate from the renewal law — producing fewer high-v steps than average.

In other words:

> *Divergence requires that the orbit spends too much time at Type A points (small $v_i = 1$) relative to the stationary distribution.*

This is a **thermodynamic** observation: a divergent orbit must lie in a low-entropy, low-dimension region of the symbolic space.

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3. The Memory Rigidity Gap (Level 2)

Level 1 shows divergence would be atypical. Level 2 asks: how atypical, and in what way?

Theorem IT2b (rigorous, given persistent-correlation condition). If the orbit's empirical process satisfies $\limsup_N |\mathrm{Cov}|*text{emp}(k_0,k_N)| > |delta > 0$, then: $$|sum*{N=1}^\infty I_\text{emp}(k_0;k_N) = \infty.$$

The memory gap. The stationary system has total memory $2\log 2$ (finite). A divergent orbit — being non-π-ergodic — would be expected to have persistent correlations, giving $\sum I_\text{emp} = \infty$.

The gap. We have not proved that divergence *implies* persistent correlations. This requires showing that an orbit cannot deviate from π without also sustaining long-range dependencies — a dynamical rigidity statement that goes beyond the Lyapunov criterion.

Stated precisely:

> *The renewal system has finite memory ($2|\log 2$ bits). A divergent orbit would carry infinite arithmetic memory. The open question is whether one can derive the infinite-memory requirement from divergence alone, without additional assumptions.*

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4. The Arithmetic Obstruction (Level 3)

This is the open frontier.

Even if Level 2 were resolved (diverge $\Rightarrow$ infinite memory), one would still need to show that infinite arithmetic memory is impossible for Collatz orbits. This is where arithmetic enters.

Theorem OE$_J$ (rigorous). For fixed truncation J: prime-starting iterates $T_\mathcal{A}^i(p)$ are equidistributed mod 2^K over primes.

Theorem OE (conditional on prime-orbit tail control PT): full fixed-time equidistribution holds.

The gap from fixed-time to individual. OE gives equidistribution *averaged over primes at each fixed step*. What is needed for the Collatz conjecture is equidistribution *along each individual orbit over time* — an orbit-level statement, not a population-level one.

The parity wall. Any individual-orbit decorrelation result would likely require controlling: $$\frac{1}{N}\sum_{n\leq N}\Lambda_C(n)\Lambda_C(n+h) \to \left(\frac{1}{N}\sum\Lambda_C\right)^2, \quad \forall h\neq 0,$$ where $\Lambda_C(n) = \mathbf{1}_{(3n+1)/2\text{ prime}}$. This is the Chowla conjecture for Λ_C, which involves controlling $\Omega((3n+1)/2)\bmod 2$. Sieve methods are known to be insensitive to this parity; no current technique breaks this barrier.

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5. The Residual Core

The four levels compress to:

$$\boxed{\text{Can one rule out infinite arithmetic memory along a single orbit?}}$$

Or equivalently:

> *Can one show $\liminf_M \frac{1}{M}\sum_{i<M}v_2(3n_i+1) > \log_2 3$ for every Collatz orbit?*

Level	Content	Status

0	Exact renewal structure; 2-bit finite memory	**Rigorous**
1	Diverge $\Rightarrow$ atypical orbit	**Rigorous**
2	Atypical $\Rightarrow$ infinite memory (under persistent-corr.)	**Conditional**
3	Individual orbit decorrelation	**Open — parity barrier**

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6. The Residual Obstruction

Within the present framework, the residual obstruction is:

> *The possibility of sustaining infinite arithmetic memory along a single arithmetic orbit.*

More precisely, the Collatz conjecture would follow if one could rule out infinite arithmetic memory along every orbit. The framework identifies three layers of what this requires:

- The **background** (renewal system) has a finite and explicitly computable amount of total memory, namely $2\log 2$.
- Non-divergent behavior is expected to remain within finite-memory regimes (consistent with the background).
- The framework suggests that divergence would require the orbit to escape into an infinite-memory regime.
- **The obstruction** to ruling this out: controlling parity correlations of $(3n+1)/2$ — the sieve parity wall.

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7. Comparison with Prior Approaches

Standard Collatz analysis focuses on:

- **Stopping times** and probabilistic drift: “on average, orbits decrease.”
- **Parity vectors** and density arguments.
- **Modular structure** and graph approaches.

These approaches capture that *generically*, orbits decrease. They do not explain why *no individual* orbit diverges.

Our framework shifts the question from:

> "Why do orbits decrease on average?" (answered by simple Lyapunov: $\log(3/4) < 0$)

to:

> "Why can no individual orbit sustain the systematic arithmetic coherence that divergence requires?" (the residual core)

The second question is harder but more precise. It identifies the parity barrier as the exact obstruction — not a technical difficulty, but a structural one.

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8. Roadmap

Paper A (Kamanoi 2026a): Establishes Levels 0 rigorously.

Paper B (Kamanoi 2026b): Establishes Level 1 rigorously; develops Levels 2–3 as a conditional framework.

The open problem: Level 3 — individual arithmetic decorrelation — is naturally connected to, and at least as hard as, Chowla-type decorrelation for $\Lambda_C(n) = \mathbf{1}_{(3n+1)/2\text{ prime}}$.

The present framework therefore suggests that the unresolved core of Collatz is not the average drift, but the possibility of orbitwise arithmetic coherence persisting indefinitely against an exact renewal background.

This note is a companion to Kamanoi (2026a–b). It does not contain new results; it organizes the logical structure of the residual problem.